

Date: 4 March

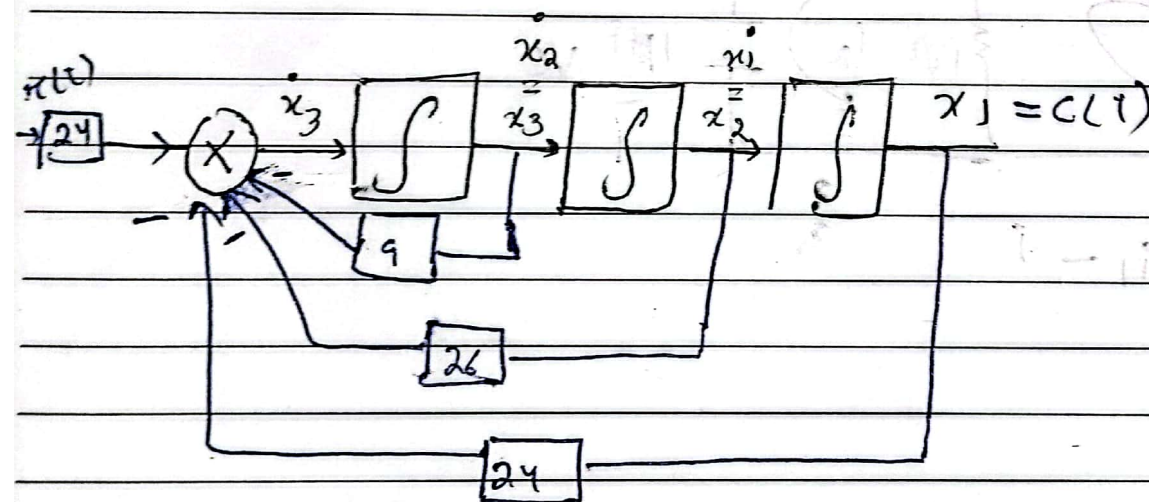
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Lecture 10

$$\frac{C(s)}{R(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} u$$

$$C = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



In diagram
 Left side
 Input right
 side output

$$\int \frac{dx_1}{dt} = x_1$$

Transfer function
 is define for
 1 input &
 1 output
 SISO & LTI

$$\frac{d^3 C}{dt^3} + 9 \frac{d^2 C}{dt^2} + 26 \frac{dC}{dt} + 24 C = 24 u$$

$$C = x_1$$

$$\frac{dC}{dt} = x_2$$

$$\frac{dx_2}{dt} = \frac{d^2 C}{dt^2} = x_3$$



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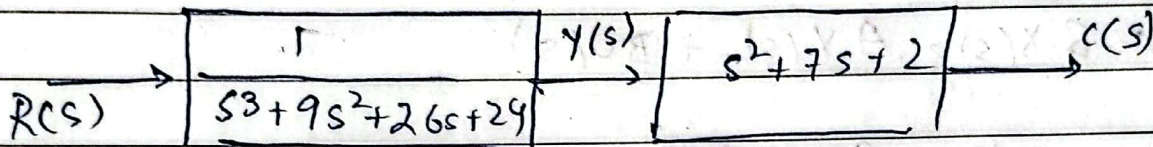
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$$C(s) = \frac{s^2 + 7s + 2}{s^3 + 9s^2 + 26s + 24}$$

$$R(s) = \frac{1}{s^3 + 9s^2 + 26s + 24}$$

1/ numerator constant
 then don't decompose

Decompose it in 2 part



$$Y(s) = \frac{1}{s^3 + 9s^2 + 26s + 24} \times R(s)$$

$$C(s) = (s^2 + 7s + 2) Y(s)$$

$$C(t) = \frac{d^2 y}{dt^2} y(t) + 7 \frac{dy(t)}{dt} + 2y(t)$$

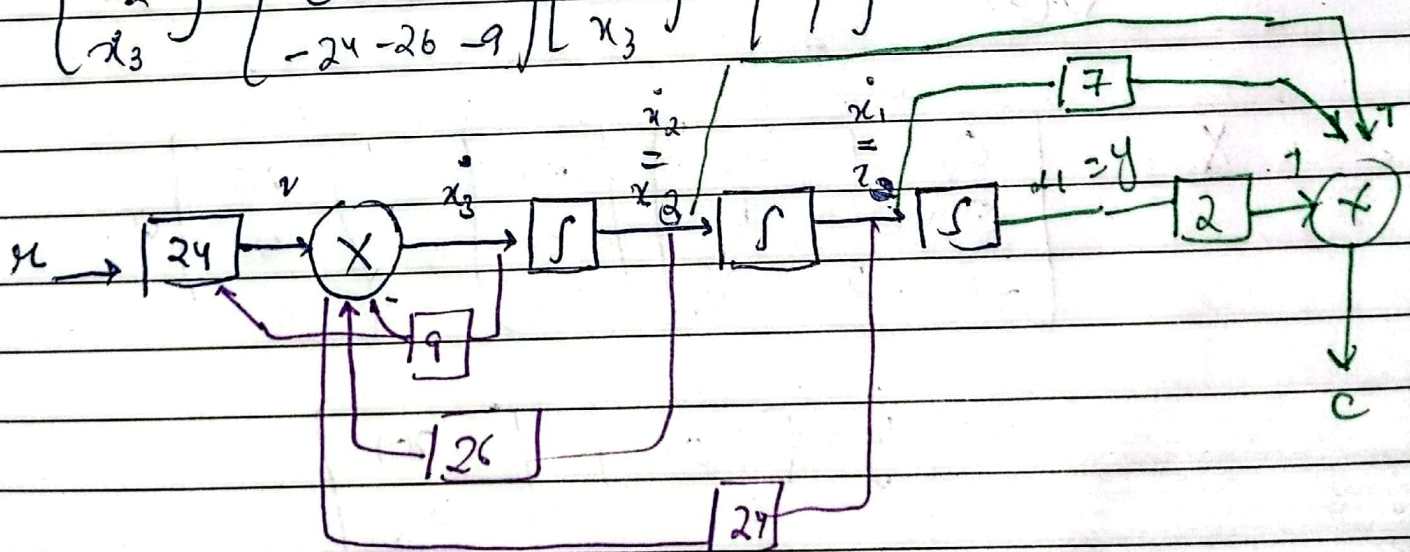
$$y = x_1$$

$$\frac{dy}{dt} = x_2$$

$$\frac{d^2 y}{dt^2} = x_3$$

$$C = [2 \ 7 \ 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$



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Converting State Space Model into TF

$$\dot{X} = AX + BU$$

$$sX(s) = AX(s) + BU(s)$$

$$Y = CX + DU$$

$$Y(s) = CX(s) + DU(s)$$

$$sX(s) - AX(s) = BU(s)$$

$$X(s)[sI - A] = BU(s)$$

$$X(s) = [sI - A]^{-1} BU(s)$$

$$Y(s) = C[sI - A]^{-1} B U(s) + DU(s)$$

$$= [C[sI - A]^{-1} B + D] U(s)$$

$$\frac{Y(s)}{U(s)} = C[sI - A]^{-1} B + D$$

$$\dot{X} = \overset{A}{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \overset{B}{\begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}} U$$

$$Y = \underset{C}{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



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$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1 & 2 & s+3 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{\text{Adj}(sI - A)}{\det(sI - A)}$$

$$\det(sI - A) = s(s(s+3)+2) + 1(0+1)$$

$$= s^3 + 3s^2 + 2s + 1$$

Adj A = [matrix of cofactors]^T

Matrix of cofactors =

$$\begin{bmatrix} s(s+3)+2 & -1 & -s \\ s+3 & s^2+3s & -(2s+1) \\ 1 & s & s^2 \end{bmatrix}$$

$$= \begin{bmatrix} s(s+3)+2 & s+3 & 1 \\ -1 & s^2+3s & s \\ -s & -(2s+1) & s^2 \end{bmatrix}$$



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$$TF = C(SI - A)^{-1}B + D$$

$$= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \frac{1}{\Delta} \begin{bmatrix} s^2 + 3s + 2 & s + 3 & 1 \\ -1 & s^2 + 3s & s \\ -s & -(2s + 1) & s^2 \end{bmatrix} \begin{bmatrix} 10 \\ 0 \\ 10 \end{bmatrix}$$

$$= \frac{1}{\Delta} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 10(s^2 + 3s + 2) \\ -10 \\ -10s \end{bmatrix}$$

$$= \frac{10(s^2 + 3s + 1)}{s^3 + 3s^2 + 2s + 1}$$